

Crash Course i AA MM5020

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Groups

Groups and subgroups

Group properties & algebraic structures Let $G \neq \emptyset$ be a set with a map $\circ : G \times G \rightarrow G$. We define the following properties (with the implicit operation $\circ$):

- **Associative:** $\forall a, b, c \in G : (ab)c = a(bc)$
- **Unit element/Identity:** $\exists e \in G : ea = a = ae$
- **Inverse elements:** $\forall a \in G, \exists b \in G : ab = e = ba$
- **Commutative:** $\forall a, b \in G : ab = ba$

Then, $(G, \circ)$ is called a

- **Semigroup** if associativity holds.
- **Monoid** if associativity and existence of a unit hold.
- **Group** if associativity, existence of a unit, and existence of inverses hold.
- **Abelian group** if associativity, existence of a unit, existence of inverses, and commutativity hold.

Uniqueness of unit and inverse Let G be a group then the following holds:

- The unit element is unique.
- The inverse of any $a \in G$ is unique.

Proof exists in KTH lecture notes.

Weak criteria for a group Let $G \neq \emptyset$ be a set with a map $\circ : G \times G \rightarrow G$ such that $\circ$ is associative:

- There exists a left unit: $\exists e \in G, \forall a \in G : ea = a$.
- All elements have a left inverse: $\forall a \in G, \exists b \in G : ba = e$.

Then $(G, \circ)$ is a group. Proof exists in KTH lecture notes.

Submonoid Let G be a monoid and $H \subseteq G$ a subset. H is a submonoid of G if

- $e \in H$
- $\forall a, b \in H : ab \in H$

Subgroup Let G be a group and $H \subseteq G$ a subset. H is a subgroup of G , written $H \leq G$, if

- $e \in H$
- $\forall a, b \in H : ab \in H$
- $\forall a \in H : a^{-1} \in H$

A trivial subgroup is the always existing subgroup $\{e\}$ alone or together with the rest of the group, since $G \leq G$. While a non-trivial subgroup is any subgroup H such that $G \neq H \neq \{e\}$ i.e the same identity element but partly the same or other extra elements.

Group homomorphisms

Let $(G, \circ)$ and $(H, \cdot)$ be two groups where $\cdot$ and $\circ$ are group operations. A function $f : G \rightarrow H$ is a group homomorphism if $\forall a, b \in G : f(ab) = f(a)f(b)$ (can be written as $G \sim H$). In other words, the function f preserves the group operation.

Meanwhile group isomorphisms are homomorphisms (same notation as above) that are bijective, i.e where no information is destroyed or created. It is written as $G \cong H$ if at least one isomorphism exists between the groups. In other words, the function f_2 preserves the group operation and is a bijection. In other words isomorphisms $\subseteq$ homomorphisms but are NOT usually subgroups.

A simple example of homomorphism is the exponential function $f : (\mathbb{R}, +) \rightarrow (\mathbb{R}^+, \cdot)$, where $f(x+y) = e^{x+y} = e^x e^y = f(x)f(y)$ but it is not an isomorphism since it is not injective (since $e^0 = e^{2\pi i}$ for example). However

the function $f : (\mathbb{R}, +) \rightarrow (\mathbb{R}, +)$, $x \mapsto 2x$ is an isomorphism since it is a homomorphism and bijective by the fact that $2x = 2y \implies x = y$ and $\forall y \in \mathbb{R}, \exists x \in \mathbb{R} : 2x = y$.

Monoid homomorphism Let G_1, G_2 be monoids and $\varphi : G_1 \rightarrow G_2$ a map. φ is a monoid homomorphism if

- (i) $\forall a, b \in G_1 : \varphi(ab) = \varphi(a)\varphi(b)$
- (ii) $\varphi(e_{G_1}) = e_{G_2}$

Group homomorphism Let G_1, G_2 be groups and $\varphi : G_1 \rightarrow G_2$ a map. φ is a group homomorphism if

- (i) $\forall a, b \in G_1 : \varphi(ab) = \varphi(a)\varphi(b)$
- (ii) $\varphi(e_{G_1}) = e_{G_2}$

Properties of group homomorphisms Let $\varphi : G_1 \rightarrow G_2$ be a group homomorphism. Then:

- (a) $\varphi(e_{G_1}) = e_{G_2}$
- (b) $\forall a \in G_1 : \varphi(a)^{-1} = \varphi(a^{-1})$

This also holds true if let $\varphi : G_1 \rightarrow G_2$ be a map between groups and $\varphi(ab) = \varphi(a)\varphi(b) \forall a, b \in G_1$ then ϕ is a group isomorphism and property b holds. OBS proof exists in KTH lecture notes.

Types of group homomorphisms

- (a) For two groups G_1, G_2 : $\text{Hom}(G_1, G_2) := \{\varphi : G_1 \rightarrow G_2 \mid \varphi \text{ is a group homomorphism}\}$.
- (b) An injective homomorphism is a monomorphism. A surjective homomorphism is an epimorphism. A bijective homomorphism is an isomorphism.
- (c) For a group G : $\text{End}(G) := \text{Hom}(G, G)$ and its elements are endomorphisms of G (i.e $\text{Hom}(Z, Z)$, where Z is arbitrary). Also, $\text{Aut}(G) := \{\varphi \in \text{End}(G) \mid \varphi \text{ is bijective}\}$ and its elements are automorphisms of G .

Composition of homomorphisms Compositions of group homomorphisms are group homomorphisms. Ex g_1, G_2, G_3 are groups and $\varphi_1 \in \text{Hom}(G_1, G_2)$, $\varphi_2 \in \text{Hom}(G_2, G_3)$ then $\varphi_2 \circ \varphi_1 \in \text{Hom}(G_1, G_3)$. Proof exists in KTH lecture notes.

Automorphism group If G is a group, then $(\text{Aut}(G), \circ)$ is a group. Proof exists in KTH lecture notes.

Kernel and image of a group homomorphism

- (a) The kernel of a group homomorphism $\varphi : G_1 \rightarrow G_2$ is $\ker \varphi := \{a \in G_1 \mid \varphi(a) = e_{G_2}\}$.
- (b) The image of a group homomorphism $\varphi : G_1 \rightarrow G_2$ is $\text{im } \varphi := \{\varphi(a) \mid a \in G_1\}$.

Kernel, image and subgroups under homomorphisms Let G_1 and G_2 be groups and $\varphi \in \text{Hom}(G_1, G_2)$.

- (a) $\ker \varphi \leq G_1$
- (b) $\text{im } \varphi \leq G_2$
- (c) φ is injective $\iff \ker \varphi = \{e\}$
- (d) φ is surjective $\iff \text{im } \varphi = G_2$
- (e) $H_1 \leq G_1 \implies \varphi(H_1) \leq G_2$
- (f) $H_2 \leq G_2 \implies \varphi^{-1}(H_2) \leq G_1$

OBS $\varphi(e_1) = e_2$. Proof exists in KTH lecture notes.

Conjugation Let G be a group and $a \in G$ an element. We define conjugation by a as $\gamma_a : G \rightarrow G$, $g \mapsto aga^{-1}$; also called an inner automorphism of G .

$$\text{Inn}(G) := \{\gamma_a \mid a \in G\} \leq \text{Aut}(G)$$

It holds that $\gamma_a \in \text{Aut}(G)$ and the function that creates all γ_a i.e $\phi : G \rightarrow \text{Aut}(G)$ where $a \mapsto \gamma_a$ is a group homomorphism (in other words $\gamma_{ab} = \gamma_a \circ \gamma_b$).

Cosets

Coset Let G be a group and $H \leq G$.

- (a) A **left coset** of H in G is a subset of the form

$$aH := \{ah \mid h \in H\}, \quad a \in G$$

- (b) $G/H := \{aH \mid a \in G\}$ denotes the set of all left cosets of H in G .

- (c) Analogously, $Ha := \{ha \mid h \in H\}$ (for $a \in G$) is a **right coset** of H in G . The set of all right cosets is denoted $H \backslash G := \{Ha \mid a \in G\}$.

The bijection $G \rightarrow G, g \mapsto g^{-1}$ restricts to a bijection $aH \rightarrow Ha^{-1}$. Hence, it defines a bijection $G/H \rightarrow H \backslash G, aH \mapsto Ha^{-1}$.

Equivalent coset conditions Let G be a group, $H \leq G$, and $a, b \in G$. Then the following are equivalent:

- (i) $aH = bH$,
- (ii) $aH \cap bH \neq \emptyset$,
- (iii) $a \in bH$,
- (iv) $b^{-1}a \in H$.

Partition of G Let $H \leq G$. Then G is the disjoint union (union where they share nothing) of all left cosets of H in G , i.e.,

$$G = \bigcup_{C \in G/H} C$$

Coset properties Let G be a group, $H \leq G$:

- (a) The elements of a coset aH in G are called the **representatives of the coset**.
- (b) $[G : H] := |G/H| = |H \backslash G|$ is the **index of H in G** . Intuitively, how many times G is bigger than H .
- (c) $\text{ord}(G) := |G|$ is the **order of G** (nr of elements in G).

Size of cosets Let G be a group, $H \leq G$, and $a, b \in G$. There is a bijection $aH \rightarrow bH$. In particular, $|aH| = |bH| = \text{ord}(H)$ (because $h \mapsto ah$ is bijective).

Theorem of Lagrange Let G be a finite group, $H \leq G$. Then:

$$\text{ord}(G) = \text{ord}(H) \cdot [G : H]$$

Normal subgroups

Normal subgroup Let G be a group, $H \leq G$. H is a **normal subgroup** of G (intuition balanced/symmetric subgroup like 0 or 1 to create group division and kernel of homomorphisms), denoted $H \trianglelefteq G$, if:

$$\forall \gamma \in \text{Inn}(G) : \gamma(H) = H, \quad \text{i.e., } \forall a \in G : aHa^{-1} = H. \quad \text{Ex } G = (\mathbb{Z}, +) \text{ and } H = (2\mathbb{Z}, +).$$

Normal subgroup criteria Let G be a group, $H \leq G$. Then the following are equivalent:

- (i) $H \trianglelefteq G$,
- (ii) $\forall \gamma \in \text{Inn}(G) : \gamma(H) \subseteq H$,
- (iii) $\forall \gamma \in \text{Inn}(G) : H \subseteq \gamma(H)$,
- (iv) $\forall a \in G : aH = Ha$.

In this case, $aH = Ha$ is called the **residue class of a modulo H** (left coset to H under a is equal to the right).

Quotient groups Let G be a group, $N \trianglelefteq G$:

- (a) $\forall a, b \in G, (aN)(bN) = abN$ (where N is not a single element but the whole normal subgroup).

- (b) $G/N = aN \mid a \in G$ is a group with multiplication as in (a), called the **factor group** or **quotient group** of G modulo N . Its unit element is $eN = N$. (Intuition: work with N instead of e and you get a group with the same structure as G but with N as the new identity element (N becomes e) and the rest of the elements are the same but with N added to them (aN instead of a , like $1*nr$)).

- (c) The **canonical projection** $\pi : G \rightarrow G/N$, $a \mapsto aN$, is a surjective group homomorphism with $\ker(\pi) = N$.

Universal property of π Let $\varphi : G_1 \rightarrow G_2$ be a group homomorphism, and let $N \trianglelefteq G_1$ such that $N \subseteq \ker(\varphi)$. Then there is a unique group homomorphism $\bar{\varphi} : G_1/N \rightarrow G_2$, such that $\varphi = \bar{\varphi} \circ \pi$, i.e., the following diagram is commutative:

Moreover:

- (a) $\text{im}(\varphi) = \text{im}(\bar{\varphi})$,
- (b) $\ker(\bar{\varphi}) = \pi(\ker(\varphi))$,
- (c) $\ker(\varphi) = \pi^{-1}(\ker(\bar{\varphi}))$,
- (d) $\bar{\varphi}$ is injective $\iff N = \ker(\varphi)$.

First isomorphism theorem If $\varphi : G_1 \rightarrow G_2$ is a surjective group homomorphism, then G_2 is canonically isomorphic to $G_1/\ker(\varphi)$.

Cyclic groups

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Group actions

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Sylow groups

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Permutation groups

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Some notation for important groups

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Rings

Rings, subrings & ideals

Ring homomorphisms

Polynomials

Integral domains

Primes

Euclidian domains

Unique factorization domains

Modules

Field extensions

Algebraic closures

Splitting fields

Finite fields

Overview of types of rings

Other good math tricks

Bijectivity and injectivity means that $f(x)=f(y) \rightarrow x=y$

Kommandon

Vandermondmatris Här är def 2

banan satsen Här är theorem nr 2 med ett *nyckelord*

linjäriseringen Här är proof nr2

Här är kommentar nr2

Här är förståelse nr2